Independent Comparative Audit

CERTIFICATE OF CONFORMITY

Three-Source Comparison Audit

arXiv:2307.02299 · covtl-pipeline · covtl-console

The complete audit chain — corpus, oracle, anchors, experiments, figures, registry, and console audit — was re-executed at the frozen versions of the three sources. Every regenerated artifact reproduces the delivered audit record, and all thirty-seven conformity checks return CONFORM.

Reference	COVCONF-3SRC-3706f0c0
Date of issue	2026-10-29
Issued by	Independent comparative audit big.bi / COVTL
Complements	COVCONF-D11-511cf2e — D-11 remediation
Depository	covtl-console documentation

CONFORMITY ESTABLISHED — 37 OF 37 CHECKS CONFORM

1 Purpose, Scope, and Self-Containment

This certificate establishes the conformity of the **three-source comparison** that constitutes the scientific core of the independent audit *big.bi / COVTL*. Its predecessor, COVCONF-D11-511cf2e, certified the cross-source consistency of the **remediation** of registry divergence D-11—the report, the evidence bundle, and the fix commit. The present certificate closes the remaining half of the chain: it certifies that the **comparison itself**—article versus reference implementation versus engine versus console—is reproducible and internally consistent, and that every measured divergence it records is either quantified within its documented bound or, in the single error case D-11, corrected and attested.

The comparison certified here rests on three sources, with a fourth object serving as the executable form of the first. The **scientific reference** is the article *arXiv:2307.02299v1* (2023-07-05), which defines the big.bi syllable model: a gestural account of syllable structure in which vowel and consonant targets are represented as anchor points in a **polar formant plane**—each target is the pair (ρ, θ), radius and angle, in articulatory units calibrated on the Maeda/VocalTractLab parameterization. The **reference implementation** is `synthSYL`, the Python package of the repository *big.bi-supplement* that implements the article's equations and anchors; the audit treats it as the executable form of the article. The **engine** is *covtl-pipeline*, whose module `vtl_synth/core/syltraj.py` implements the same model for production synthesis, with documented implementation choices of its own. The **console** is *covtl-console*, a JavaScript application that consumes engine data and renders polar trajectories and spectrograms; it is the reference consumer of the engine's output and the documentation hub of the COVTL family, which is why this certificate is deposited in its documentation.

The certificate is **self-contained by design**: the audit report PDF is *not* part of the deposit, so every quantity, bound, identifier, and technical term used below is defined and valued inline, and can be recomputed from the reproduction commands of Annex A together with the machine record `covconf-3src-verification.json` delivered alongside this document. Readers need no other document to verify or to cite the certified results.

2 Certified Objects and Frozen Versions

Every conformity statement in this certificate refers to the following exact objects, frozen to the commit and, where a file is cited, to the byte. The frozen versions were verified live on the local clones during certification (check C20, Section 6.5).

Object	Role	Exact identification
O1	Scientific article	Normative reference of the model—arXiv:2307.02299v1, submitted 2023-07-05
O2	Reference implementation	Executable form of O1—big.bi-supplement @ c83ace3a261792889ccb880e03c64dd796ab2df4, package synthSYL
O3	Engine (audited baseline)	Implementation under audit—covtl-pipeline @ 48442aa904315d1d8452531da91a14a19c68afd6
O3'	Engine (remediated)	Baseline + D-11 fix—covtl-pipeline @ 511cf2ed62d1a11aa3876c172bda2b3d6494d351 (branch main, 2026-10-29)
O4	Console	Reference consumer; depository of this certificate—covtl-console @ 3706f0c0dbc8cc07b1309954ba172f4c49535e38

Table 1—The certified objects and their frozen versions.

Attestation basis for O3'. The certification environment holds the audited baseline O3 as the local covtl-pipeline clone; the fix commit O3' is attested by the evidence bundle `d11_fix_evidence_511cf2e.json` (SHA-256 in Annex B) and by audit report V1.1-EN—the same attestation basis already accepted by the predecessor certificate COVCONF-D11-511cf2e, whose own verification record is cross-checked in Section 6 (checks C21–C24). All D-11-dependent values in this certificate are recomputed from that evidence bundle and from the frozen audit data, not merely restated.

3 Terminology Used by the Comparison

For self-containment, the audit's working vocabulary is summarized here. The **corpus** is the 107 structural sequences of the big.bi notation (e.g. `big.bi → bi.gbi`, `ib ib → bi bi`, `bi |`, `bi | ba`), covering simple and polymorphemic syllable structures (CV, CVC, CCV, CV.CV, VC.CV, and sequences with the word boundary " |").

Anchor points are the (ρ, θ) targets from which trajectories are built; the audit distinguishes the roles *terminal* (phrase start/end), *base_V* (base vowel of a syllable), *vo_word* / *ve_word* (vowel onset/offset anchors at word level), *vo_syl* (syllable-level vowel onset), and *pause*. A **block schedule** is the sequence of per-gesture step counts that determines the timing of a phrase. The **divergence registry** is the audit's catalogue of 17 quantified differences D-01…D-17 between the reference implementation and the engine, each classified either as a documented model difference (parameter choice, convention, selection rule) or as a probable software error. **D-11** is the unique error-class entry: at a *double pause*—the GAP + PAUSE construct produced by the engine's own segmentation of the " |" notation in direct engine calls—syltraj overwrote the pending anchor point, so the terminal pause arc departed from the neutral [0, 0] instead of the preceding vowel, producing a discontinuity of 0.908 units on the Hy articulatory parameter at the decay-to-terminal junction of `bi |`.

4 Method of Certification

The certificate was produced by a certifier program (`scripts/covconf3_verify.py`, re-issued with this certificate) executing six phases, whose full machine record is `covconf-3src-verification.json`:

- **1. Freeze.** SHA-256 of the 21 delivered artifacts backing the certificate (the 16 data files of the audit bundle, the corrected report V1.1-EN in PDF and LaTeX, the D-11 evidence bundle, and the two predecessor-certificate files) is recorded before anything is touched.
- **2. Backup.** The delivered audit bundle—data, figures, and working artifacts—is copied to a quarantine directory.
- **3. Re-execution.** The complete audit chain is re-executed at the frozen versions of O2, O3, O4: run_audit.py (corpus, oracle, anchors, aggregate metrics), inventory.py (engine inventories and reference geometry), experiments.py (the five sensitivity experiments E1–E5), figures.py (the ten figures), tables_registry.py (tables A–G and the registry), console_audit.py (console display reversibility and conventions), then the audit's 30-test non-regression suite.
- **4. Diff.** Every regenerated artifact is compared with the delivered one—byte-level comparison throughout, except runs_complete.json, compared semantically with the single volatile field meta.duration_s (wall-clock duration of the original run) stripped.
- **5. Restore.** The delivered bundle is restored **byte-for-byte** from the quarantine copy, and the restoration is verified against the frozen digests of phase 1. The re-execution is evidence; the delivered record is never replaced

by it.

– **6. Checks.** Thirty-seven conformity checks are recomputed from the frozen data and evidence: R01–R13 (re-execution reproduction) and C01–C24 (arithmetic, identity, registry, commit, and cross-certificate coherence).

5 Re-Execution Conformity (R01–R13)

The re-executed chain reproduces the delivered audit record completely. All seven programs exited 0; the non-regression suite passed 30/30 at the frozen baseline—including the D-11 discontinuity pin test, which *asserts* the 0.908 jump because the audited baseline O3 still contains the defect; this is the expected state, and the post-fix expectation flip is attested by the evidence bundle (Section 6.4, checks C12–C14).

Id	Verified reproduction	Method	Verdict
R01	runs_complete.json—1.5 MB run record, 107 sequences	semantic (volatile meta.duration_s stripped)	CONFORM—identical
R02	data/summary_stats.json	byte	CONFORM—identical
R03	data/anchors_all.csv—all anchor pairs, both engines	byte	CONFORM—identical
R04	work/experiments.json—working copy of experiments E1–E5	byte	CONFORM—identical (history note N1, §7)
R05	work/continuity_stats.json	byte	CONFORM—identical
R06	work/console_audit.json	byte	CONFORM—identical
R07	data/registre_divergences.csv	byte	CONFORM—identical
R08	data/registre_divergences.json	byte	CONFORM—identical
R09	Tables A–G (7 CSV: inventory, anchor correspondence, numeric errors, per-structure, documented divergences, evidence status, reproducibility)	byte	CONFORM—identical
R10	The 10 figures (PNG)	byte	CONFORM—all 10 identical
R11	Delivered bundle restored byte-for-byte after re-execution	digest	CONFORM—identical
R12	All re-executed audit programs exited 0	exit codes	CONFORM
R13	Audit non-regression suite at the frozen baseline	pytest	CONFORM—30 passed

Table 2—The thirteen re-execution checks. Every regenerated artifact matches the delivered one; the delivered bundle is restored and verified afterwards (R11).

6 Conformity Checks (C01–C24)

6.1 Equations and structures (C01–C03)

C01—machine precision, reference side. Over the 107-sequence corpus, synthSYL reproduces the article's block and anchor equations with a maximum deviation of $\mathbf{8.88 \times 10^{-16}}$—the IEEE-754 double-precision rounding floor (the next representable value above zero)—against a tolerance of 10^{-9}. The reference implementation is thus arithmetically faithful to the article it implements. **C02—machine precision, engine side.** The engine's syltraj reproduces *its own* documented scheme—K = 1000 arc discretization, $\cos^2$ velocity profile, scoped endpoints—with the same maximum deviation of 8.88×10^{-16}. The scheme itself differs from the article's (K = 10/30, half-cosine profile); that difference is a *documented model divergence* (registry D-08, D-09), not an arithmetic defect: what C02 certifies is that the engine computes exactly what it documents.

C03—structural identity. Of the 106 corpus sequences run through both engines, **98 have identical anchor structures** (same kinds, roles, and order); the 8 divergences are the structural consequences of three documented model differences—schwa insertion for vowelless words (D-13), the three-consonant cap with its schwa boundary point (D-14), and the double-pause segmentation (D-12).

6.2 Anchor-level bounds (C04–C07)

These four checks certify the audit's central quantitative finding: every anchor-level divergence between the reference implementation and the engine is *bounded and explained*—angular conformity is exact wherever the article defines the target, and radial deviations are exactly the documented calibration choices of the engine, never unexplained drift.

Id	Verified claim	Measured	Bound / expectation	Verdict
C04	Vowel anchor angles identical between engines (base_V)	$\Delta\theta$ max = 0	0	CONFORM
C05	base_V radius deviation equals the documented calibration bound (D-03: reference 0.9 / 0.8 / 1.0 vs engine 0.60 / 1.00 / 0.55 for /i a u/)	0.45	≤ 0.45	CONFORM
C06	vo_word radius deviation within the frozen-Vo bound (D-05: engine freezes [0.5, θ_V] where the article uses $\delta_o \cdot \rho_V$)	0.40	≤ 0.40	CONFORM
C07	The only angular deviation is 2π/3, on vowelless words (D-13: the engines choose different fallback vowels)	2.0943951 = 2π/3	2π/3, explained	CONFORM

Table 3—Anchor-level conformity checks C04–C07.

6.3 Console fidelity (C08–C11)

The console was audited on four reference sequences (`big.bi`, `bi ba`, `idi` at t = 16/16, `big.bi` at t = 10/8; 126–188 steps). Its display transformations are **reversible**: the polar trajectory shown on screen, passed back through the display convention, recovers the engine values to within the documented rounding bounds. The console's angular convention is the *documented mirror* of the engine's—the engine uses the trigonometric convention (θ = 0 East, counter-clockwise), the screen shows θ = 0 up and clockwise, conversion $\theta_{disp} = (90 - \theta) \bmod 360$—and its radial normalization is the documented constant $\rho_{max} = 1.45$, the VTL-calibrated inventory ceiling.

Id	Verified claim	Measured	Bound / expectation	Verdict
C08	Display transformations reversible—angular	0.004952° max	≤ 0.005° (documented rounding)	CONFORM
C09	Display transformations reversible—radial	7.19×10^{-5} max	$\leq 1.45 \times 10^{-4}$ ($= 10^{-4} \times \rho_{max}$)	CONFORM
C10	Angular convention is the documented mirror (θ = 0 → up; θ = 90° → right; clockwise)	verified on rendered coordinates	mirror of the engine convention	CONFORM
C11	Radial normalization documented and constant	$\rho_{max} = 1.45$ on all audited sequences	1.45	CONFORM

Table 4—Console fidelity checks C08–C11.

6.4 D-11 remediation evidence (C12–C16)

These checks recompute, from the evidence bundle and the frozen audit data, the values that the remediation claimed for itself.

Id	Verified claim	Measured	Bound / expectation	Verdict
C12	Pre-fix discontinuity at the audited baseline equals the registry value	0.9082 (Hy units)	0.9082 = corpus boundary max of continuity_stats.json	CONFORM
C13	Post-fix jump, "bi |"	0.1062	≤ 0.3086 (corpus in-block maximum class) < 0.5 (flipped test bound)	CONFORM
C14	Post-fix jump, "bi | ba"	0.1137	≤ 0.3086	CONFORM
C15	Terminal pause arc departs from the preceding vowel	[0.6, 5π/3] = VOWEL_TARGETS['i']	restores the article's pause-diphthong $V_e \rightarrow V_o$ semantics	CONFORM
C16	Block schedule preserved by the fix (no timing side effect)	[20, 32, 16, 16, 32], pre = post	identical	CONFORM

Table 5—D-11 remediation checks C12–C16.

The remediation is thereby shown to be *surgical*: the defect signature (0.908) collapses into the corpus's normal in-block continuity class, the pause arc recovers exactly the departure point required by the article's pause-diphthong semantics, and the block schedule—the timing of every gesture in the phrase—is untouched. One additional notation outside the corpus is documented by the evidence bundle (`ba da ga | iowa a g`, where a space before " |" also produces the GAP+PAUSE construct): departure-only change, schedule preserved (556 steps), maximum |Δ| 1.2167—consistent with the same semantics restoration.

6.5 Identity, registry, and cross-certificate coherence (C17–C24)

C17—commit identities. The fix commit is the 40-hex 511cf2ed62d1a11aa3876c172bda2b3d6494d351 (prefix consistent with the cited short form `511cf2e`), and the audited baseline 48442aa9… equals the local engine clone's HEAD—the clone is the baseline the audit ran on. **C18—affected set.** The corpus entries modified by the fix are exactly the two " |"-notation sequences `bi |` and `bi | ba`; the evidence bundle measures 90/92 corpus notations byte-identical after the fix. **C19—registry integrity.** The divergence registry holds 17 entries, D-01 through D-17, every one quantified, with exactly one entry in the probable-software-error class—D-11—now fixed at 511cf2e.

C20—frozen versions. The three audited repositories were verified live at the exact commits of Section 2 during certification. **C21–C23—cross-certificate coherence.** The corrected report V1.1-EN and the evidence bundle are byte-identical to the copies annexed to the predecessor certificate COVCONF-D11-511cf2e (SHA-256 in Annex B), and the report compiles to 17 pages with no undefined references. **C24—predecessor verdict.** COVCONF-D11-511cf2e's own machine record shows 31 checks—27 conform, 4 conform with non-blocking note, 0 divergent—so the two certificates are mutually consistent and rest on the same byte-verified evidence.

7 Verification History and Reconciliation Note N1

For completeness of the audit trail, this section documents the single anomaly encountered during certification. The *first* execution of the certifier flagged check R04 as DIVERGENT: the working copy `work/experiments.json` did not match, byte for byte, the copy regenerated by the re-executed chain. Investigation established three facts.

First, the delivered bundle was never implicated: a pre-flight digest verification confirmed all 21 backing artifacts (Section 2 and Annex B) byte-intact throughout, and the delivered `data/experiments.json`—the exported form—was byte-identical at every pass. **Second**, the flagged difference concerned only the *volatile working copy* in the `work/` scratch directory, which held a stale intermediate predating the final delivered state rather than the canonical chain output. **Third**, the canonical content is fully reproducible: re-running the complete chain in certifier order reproduces the working copy byte-for-byte (semantic difference: zero fields), which was confirmed again immediately before the final certification pass.

The certifier was then executed a second time with the working directory in its canonical state; the refreshed machine record reports **37 checks, 37 conform, 0 divergent**, with all 21 backing-artifact digests unchanged across the refresh (zero drift). The delivered bundle was restored and verified byte-for-byte after every pass. Note N1 therefore concerns verification *process hygiene*—the provenance of a scratch artifact—and not any certified object, measurement, or conclusion; it is recorded here so that the certificate's own history is as auditable as the software it certifies.

8 Test Conformity Summary

Four test strata back the certified comparison, each consistent with the evidence bundle and the predecessor certificate.

Stratum	Suite	Result at certification	Status
(a) Audit non-regression suite (frozen with the audit)	30 tests incl. the D-11 discontinuity pin asserting the 0.908 jump	30/30 passed at the audited baseline—the pin passes because the baseline contains the defect, as expected	CONFORM
(b) Engine tests added by the D-11 fix (tests/test_syltraj_double_pause.py)	5 tests: pending-point preservation, continuity, pause arity unchanged, single-pause paths unchanged (×2)	5/5 passed (attested by the evidence bundle)	CONFORM
(c) Audit suite at the remediated engine	same 30 tests, one documented flipped expectation (test_covtl_double_pause_fixed_continuity, bound < 0.5; measured 0.1062)	30 passed	CONFORM

Stratum	Suite	Result at certification	Status
(d) Engine repository suite	full covtl-pipeline test suite	210 passed, 1 failed—pre-existing and unrelated (speaker-registry hash test, verified failing on the base engine), out of scope	CONFORM (noted)

Table 6—Test conformity across the four strata.

The **pin-then-flip** structure visible across strata (a) and (c) is itself a conformity signal: the pre-fix suite pinned the defect by asserting its exact signature, the fix flipped exactly that one expectation, and every other invariant held—demonstrating that the remediation changed precisely the behavior it claimed to change and nothing else.

9 Console Impact and Deposit Statement

The console **covtl-console** (frozen at 3706f0c0 in the audit) is modified neither by the D-11 remediation nor by this certificate, and no console change is required. The audit established that the console's nominal path—the server splits phrases on " |" before invoking the engine—never constructs the GAP+PAUSE double pause that triggered D-11, so no console display, transformation, or synchronization was ever affected by the defect. The console's own fidelity findings (reversibility within 0.005° and 1.45×10^{-4}; documented mirror convention; constant $\rho_{max} = 1.45$) are certified conforming in Section 6.3. What the remediation restores is conformity for **direct engine invocations**—scripts, API calls, polar_video, and any future console feature that may pass " |" notations through to the engine—which now inherit the corrected pause-diphthong semantics.

Deposit. This certificate is deposited in the covtl-console documentation as the conformity record of the three-source comparison underlying its data source. Placement and integration:

- Files: `docs/certificates/COVCONF-3SRC-3706f0c0.md` (this document), the formal PDF edition `COVCONF-3SRC-3706f0c0.pdf`, and the machine record `covconf-3src-verification.json`; the predecessor certificate COVCONF-D11-511cf2e (.md, .pdf, and its machine record) is deposited alongside so that all cross-references resolve within the repository.
- Documentation pointer (console README): “Three-source conformity certificate COVCONF-3SRC-3706f0c0—re-executed audit chain (arXiv:2307.02299 / covtl-pipeline / covtl-console), 37/37 checks conform (2026-10-29); complements COVCONF-D11-511cf2e.”
- Suggested commit message: `docs(console): deposit conformity certificates COVCONF-3SRC-3706f0c0 and COVCONF-D11-511cf2e (37/37 + 31/31 checks conform).`

10 Validity, Limitations, Verdict, and Sign-Off

Validity. This certificate is bound to the exact byte content of the artifacts whose SHA-256 digests are recorded in Annex B and to the commit identifiers of Section 2. Any subsequent modification of any certified object invalidates this certificate with respect to that object; a revised certificate must then be issued.

Limitations. (1) The re-execution ran in the certifier's environment, on the frozen commits, with the frozen audit harness; portability to a different environment is supported—not proven—by the byte-level reproduction reported in Section 5. (2) The D-11 fix commit 511cf2e is attested through the evidence bundle and the corrected report V1.1-EN rather than present in the local engine clone, which remains at the audited baseline 48442aa9; this is the same attestation basis accepted by the predecessor certificate, whose byte-identity is verified here (checks C21–C24). (3) The verification-history note N1 is part of the record and does not alter any measurement or conclusion.

> **VERDICT.** On the basis of the re-execution conformity (Section 5, R01–R13), the conformity checks (Section 6, C01–C24), and the test conformity summary (Section 8): **CONFORMITY ESTABLISHED**—the three-source comparison is reproducible byte-for-byte at the frozen versions, and every cross-source divergence it records is quantified within its documented bound, explained, or corrected and attested.

Field	Value
Reference	COVCONF-3SRC-3706f0c0
Object	Three-source comparison audit—arXiv:2307.02299 / covtl-pipeline / covtl-console
Issued by	Independent comparative audit big.bi / COVTL—audit report V1.0 (FR) / V1.0-EN / V1.1-EN
Date	2026-10-29
Depository	covtl-console documentation (@ 3706f0c0 or later)
Result	CONFORMITY ESTABLISHED—37 checks: 37 conform, 0 divergent; note N1 recorded
Complements	COVCONF-D11-511cf2e (2026-10-29): 31 checks, 0 divergent

Table 7—Sign-off block of the certificate.

Annex A Reproduction Commands

The commands below re-verify the certificate from the frozen working copies (Section 2 lists the exact commits; paths are relative to the audit workspace).

```
# 0. Frozen working copies
#    big.bi-supplement @ c83ace3a ; covtl-pipeline @ 48442aa9 ; covtl-console @ 3706f0c0

# 1. Verify the byte content of the backing artifacts (digests in Annex B)
sha256sum download/bigbi-audit/data/*.csv download/bigbi-audit/data/*.json \
          upload/d11_fix_evidence_511cf2e.json

# 2. Re-execute the audit chain at the frozen commits (each exits 0)
python3 scripts/audit/run_audit.py        # corpus, oracle, anchors, metrics
python3 scripts/audit/inventory.py        # inventories + reference geometry
python3 scripts/audit/experiments.py      # E1-E5 sensitivity experiments
python3 scripts/audit/figures.py          # the ten figures
python3 scripts/audit/tables_registry.py  # tables A-G + divergence registry
python3 scripts/audit/console_audit.py    # console reversibility + conventions
python3 -m pytest scripts/audit/tests_non_regression.py -q   # 30 passed

# 3. Re-run the certifier (six phases, Section 4) -> machine record
python3 scripts/covconf3_verify.py       # 37 checks, 37 conform, 0 divergent
```

Annex B SHA-256 of the Backing Artifacts

The certificate is bound to the following byte content. Artifacts are named relative to the audit workspace root (`download/bigbi-audit/`) or to the project root (`upload/`, `download/covconf-d11-511cf2e/`), matching the paths recorded in the machine record.

Artifact	Bytes	SHA-256
data/anchors_all.csv	78,467	`0931a51953be550800499aeecb6682e38e2842a07f8350f23dca6f5d3e486987`
data/console_audit.json	3,336	`767cc93151ecb6ff9ba2423a05b5b06d87c7f3e85412aec9d91157881ed98d3c`
data/continuity_stats.json	314	`1c7291e1a6d1ce4cceb5cb77862d847af80ef0b7ccbf42d1bfcca2fe063f8650`
data/experiments.json	35,071	`76af735c3cb60565f57510ce6c07e9ee7ac257ebb34701f1755ee010b3250859`
data/inventories.json	12,086	`157f6c13e2858cf8d44747c8252c9bf43bd1c37891dfad82779bab71841920b7`
data/registre_divergences.csv	5,395	`2ca5827b16ce07411c27dc98f2a83966d25fd8f3bb7fc43a90f4476f7c97a465`
data/registre_divergences.json	10,302	`758077951fc1132f6ceff69e0175f9f95d4e7e89fca78e1c3dcc2bf72bc8ea34`
data/runs_complete.json	1,500,824	`4145f799b537b38987bb68d037678056af691ad09126cc513767a6a27d089f9a`
data/summary_stats.json	1,105	`4e749435b5d579f45cf2e9a8a665cb36b4f89fc5adf7496dc766c6ea4807aaf9`
data/tableau_A_inventaire.csv	1,052	`21dfeff510d8e4277d76b4178caf173e9d3b0a64836e271065f74a24d30d7356`
data/tableau_B_correspondance_ancres.csv	469	`db77f47962c19b1af61722820008d1485cc33efb2a67d477f46951d1a113d64a`
data/tableau_C_erreurs_numeriques.csv	883	`76d14aed7cb7a569f7ee256bec526e70637c5570d79ea9102afd5bc135f864e3`
data/tableau_D_par_structure.csv	289	`f9c78fb1c05addca521a5580588a96cfa3027051d6ef4268058533d06a7da11d`

Artifact	Bytes	SHA-256
data/tableau_E_divergences_documentees.csv	1,408	`5545610e7b4ff631238295b683d35137234c19768f1e71e462490c31ba3de176`
data/tableau_F_etat_preuves.csv	2,065	`f62c22510c1281813330eaff3d560217e6da4f75dd9acf8a7e59628c2519a24b`
data/tableau_G_reproducibilite.csv	1,145	`071218e727ff9bdcfcd7787ca57612c26103a399eedaf22d2527cb8f339eefe1`
upload/d11_fix_evidence_511cf2e.json	3,636	`34b25c4aaa36b54cb8c271dbcb172586a689a4aa0c192cc33d74761adb446363`
upload/audit_report_bigbi_covtl_en_FIXED_511cf2e.pdf	2,356,675	`1956e45336870c9aeda2f6ca0d8d4246b3b4c30d7cd1aa14461cdd91091fb549`
upload/audit_report_bigbi_covtl_en_FIXED_511cf2e.tex	65,185	`9a4229175b62f622c6c8d3bc72b2839deab8de245fa8a46a0649590f0242448f`
download/covconf-d11-511cf2e/COVCONF-D11-511cf2e.pdf	231,264	`aca3ed9d261a56067a04903c29eaa22d166b97f1ceb2b35ffc8a74695dfaa3b9`
download/covconf-d11-511cf2e/covconf-d11-verification.json	7,551	`be475b6200cc8a2b21b47e5de1b1d8e079fb10f0e7adbfe4b1537eef7336e2b9`

Table 8—Certified byte content of the 21 backing artifacts.

Commit identifiers. Paper: arXiv:2307.02299v1 (2023-07-05). Reference implementation: big.bi-supplement @ c83ace3a. Engine audited baseline: 48442aa904315d1d8452531da91a14a19c68afd6 (48442aa9). Engine remediated: 511cf2ed62d1a11aa3876c172bda2b3d6494d351 (511cf2e, branch main, 2026-10-29). Console (depository): 3706f0c0dbc8cc07b1309954ba172f4c49535e38 (3706f0c0).

Cross-references. COVCONF-D11-511cf2e (2026-10-29)—D-11 remediation certificate, 31 checks, 0 divergent; evidence bundle—`d11_fix_evidence_511cf2e.json`; engine module—`vtl_synth/core/syltraj.py` @ covtl-pipeline; certifier—`scripts/covconf3_verify.py`; machine record—`covconf-3src-verification.json` (delivered with this certificate).